

CANDY DIALOGUE ENGINE

Variants & Translations

1.1.0

What should you do if you want to translate your dialogues to other languages, or create alternate texts for a same spoken line? Candy DE offers an elegant solution: variants.

Creating Variants

In the Candy Dialogue Creator, open the settings menu (⚙️ button in the top-left corner), then click the "Variants" tab:



Adding Languages

To add a language, click the "Add Language" button and type a language name. Spaces and underscores (_) can't be included in language names. If you wish to create multiple variations of the same language, we suggest using dashes (e.g. 'English-US' or 'English-UK').

Each language can be assigned a default text direction by clicking the arrow button to the right, in the list.

Adding Variants

To add a variant, click the "Add Variant" button, and type a name. Variant names can't include spaces, and underscores must only be used to prefix tags that Candy DE should account for when selecting a variant to display (see 'Tags' below).

The pound sign (#) should only be used for randomness tags (see Randomness Tags below).

Note that when a new variant is created, it is added to every language. Likewise, when a new language is added, every variant is added to it.

Attribute Tags

Candy DE is capable of automatically selecting a variant to display based on language and variant tags.

By default, two tags are supported: '_S:' for speaker gender, and '_P:' for player gender. You can add support for your own custom tags in Candy DE as needed.

Custom tags must always match the following format:

`_[string]:[value]`

For example, for a variant that displays when the speaker is female, you would use the following tag:

`_S:F`

(replace 'F' with the value you use for the female gender in your game)

A variant that displays when the speaker is female and the player is male would be named like this:

`_P:M_S:F`

(replace 'F' and 'M' with the values you use for gender in your game)

Tag order in variant names does not matter: Candy_DE will check all tags, and will only use variants if all their tags match. To avoid accidentally creating duplicate variants (i.e. variants with the exact same tags and values, but in different orders) we suggest always listing tags in alphabetical order inside variant names.

Note that tag names don't have to be a single letter. For example, if you want a tag for the player's class, you could create a variant '_P-Class:'. Candy DE will use the underscores to identify where tags begin and end, and it will use the colons (:) to separate tag names from their values.

Randomness Tags

Candy DE can select at random between multiple variants. Simply create copies of a variant, with the added '_#[int]' tag at the end of the name. For example:

_S:F
S:F#1
S:F#2
S:F#3

→ **One of these will be selected at random.**

The non-random variant must exist (e.g. '_S:F'), and it will be included in the random selection. The non-random variant must also have text inside the spoken line. Otherwise, the random variants will never have a chance to display (if it helps, think of the non-random variant as _#0).

Variants can be assigned weights to increase or decrease their chances of being selected relative to one-another (see the Writer UI section further below).

Any [int] will work for random tags, and the numbers don't need to follow one-another. For example, if you have '_S:F_#1' and '_S:F_#3', but '_S:F_#2' is missing, this will not cause errors.

TTS Tag

If you use Candy DE's Text-To-Speech (TTS) compatibility features, you have the option of creating special TTS versions of each variant.

When using these TTS versions, the normal variant (e.g. "Default_S:F_#1") is displayed on screen, while a matching TTS variant is passed to your TTS tool.

You can therefore write the TTS version in a way that will produce better tone, pacing or pronunciation from your TTS tool, or even use special tags and syntax that your TTS tool might support. For example, display "Candy DE" on screen, but write it as "Candy Dee E" for the TTS tool to avoid pronouncing it "Candy Deh".

To create a TTS variant, simply add the '_TTS' tag at the very end of the variant name (after randomness tags), like so: "Default_S:F_#1_TTS".

Note that TTS requires some setup: see the [TTS.pdf](#) guide for more information.

Candy DE

Behavior

When a Spoken Line has multiple variants, Candy DE selects a variant to display based on the following criteria:

Step 1 - Initial Selection:

- Variants without any text are discarded (Candy DC also does not include empty variants when exporting).
- Variants are retained if all tags have their condition fulfilled. For example, **French_P:M_S:F** will only display if the language is set to French, the player's gender is "M" and the speaker's gender is "F".
- Randomness tags are ignored until Step 4.

Step 2 - Cleanup:

Variants are discarded if all their tags exist in another variant, and if that other variant also has more tags.

For example, if a Spoken Line has the variants **Default_P:M_S:F** and **Default_P:M_S:F_A:G**, **Default_P:M_S:F** will be discarded.

Step 3 - Ties:

At this point, multiple variants may remain if they have different tags. For example, **Default_P:M_S:F_A:G** and **Default_S:F_C:W**.

If multiple variants remain, one will be selected at random. The weight of each variant will be taken into account.

Step 4 - Randomness Tags:

At this stage, a single variant has been selected (for example, **Default_P:M_S:F_A:G**).

If variants with randomness tags exist for this variant (e.g. **Default_P:M_S:F_A:G_#1** and/or **Default_P:M_S:F_A:G_#2**), one of them will be selected at random, based on each variant's weight.

Remember that the original variant (e.g. **Default_P:M_S:F_A:G**) will be part of the random selection as well.

Step 5 - No Suitable Variants:

If no suitable variant is found throughout this process (i.e. no variant has text and all of its tags fulfill their condition), the line is skipped. No visible notice is provided to the player, so this is completely discreet.

This can be helpful: for example, in languages where sentences tend to be longer, you can split the text over multiple lines if space in the dialogue box is limited; this will not affect on-screen performance for other languages.

Note that **_TTS** variants are handled elsewhere and are not part of the above logic.

Settings

Several exported variables in the Candy_Engine script can be used to configure how variants are handled:

random_variants (bool):

If set to 'false', Step 4 above will be skipped (random variants won't be used).

mp_count_variants (bool):

This variable determines whether or not skipped lines are counted towards §Media_Pause commands, which can be configured to make a media to resume once a specific number of lines have been processed.

Setting this variable to 'true' will ensure dialogue always resumes at the exact same point, even if some Spoken Lines were skipped. Setting this variable to 'false' makes media resume based on the number of Spoken Lines that were actually display on screen.

Alternately, you can also work around that by using a variable reference for the number of lines to count inside §Media_Pause commands, and adjusting the variable value based on language.

player_tags (dict) & speaker_tags (dict):

These dictionaries can be modified to add custom tags that you want Candy DE to handle. For keys, provide a tag name. For values, provide the actor property that should be tested (in the **actors** dictionary).

For example, "S": "Gender" will lookup the actor's "Gender" key in the **actors** dictionary.

Note that the **player_tags** dictionary requires that you assign an actor to **current_player** in **Candy_Database.gd**, which allows the engine to know which actor's data it must look up for the player.

passed_tags (array) :

This array indicates any tags that automatically succeed their condition. In other words, passed tags will never prevent variants from being selected.

Note that this is not an automatic selection for variants: variants can still fail to be selected if other tags don't fulfill their condition, or due to other factors.

Remember that you can append or remove tags from this array at runtime.

disabled_tags (array):

This array indicates any tags that automatically disqualify a variant from being selected. In other words, disabled tags will be treated as if their condition has failed, even if it actually passes, and variants featuring these tags will never be selected.

Remember that you can append or remove tags from this array at runtime.

Note that any tags present in both `disabled_` and `passed_` arrays will pass (pass takes priority).

Language

Language should be set in `Candy_Properties.gd`, in the `language` variable.

This should be done through code, at runtime, e.g. in an option menu where the player can select their preferred language.

Dialogue engine instances can also use their own language setting: this is done through the `engine_language` variable, and it will override the `language` variable in `Candy_Properties.gd`.

Candy DC Spoken Lines

Variant List

When editing Spoken Lines, the list of languages and variants is displayed to the right of the writing/preview area.

A variant can be selected for editing by clicking on it. Variants written in white have text written to them, while variants in yellow are empty.

In the list, the number on the right of each variant represents their assigned weight.

Variant Data

Weight can be modified in the Weight field below the writing area.

For each line, if the default text direction of a variant is not desirable, a specific text direction can be provided in the Text Direction field below the writing area.

When exporting dialogues, empty variants in any Spoken Line are not exported to avoid unnecessary bloat.

Voice Preview

Pressing the 'Play' button for previewing voice files will play the file for the currently selected variant, if such a file exists.

As a reminder, voice files for the "Default" variant must be placed in the following location:

`Candy_DE/Media/Characters/Voices/[speaker]/[conversation]/[block]`

Voice files for all other variants must be placed in the following location:

Candy_DE/Media/Characters/Voices/[speaker]/[conversation]/[block]/[variant name]

Translation Area

The translation area can be displayed by toggling the "Translation" button at the top of the data area.

The translation area can be used to view the text for a different variant, which is ideal when translating from one language to another. The text is shown raw (no BBCode tags visible) and with BBCode formatting.

The translation area displays the number of words and characters in the displayed variant, as well as the word and character difference with the variant currently being edited.